

ARJUN VARMA

(347) 987-9427 | av3342@columbia.edu | linkedin.com/in/varma-arjun/ | github.com/ARJUNVARMA2000 | arjun-varma.com

EDUCATION

Columbia University

New York, NY

Master of Science in Data Science

Dec 2026

Coursework: Applied Machine Learning, Agentic AI for Analytics, Statistical Inference and Modeling, Probability and Statistics

Teaching Assistant, Columbia Business School: Business Analytics II (Foundations of AI) and Hollywood and Big Data

Vellore Institute of Technology

Vellore, IN

Bachelor of Technology in Electronics and Communication Engineering

May 2022

Special Achiever Award | Merit Scholarship

WORK EXPERIENCE

Novo Nordisk

Plainsboro, NJ

Data Scientist Intern

Jun 2026 - Aug 2026

- Improving next-best-engagement models and developing an LLM layer that identifies key HCP drivers and generates tailored field guidance across access, efficacy, competitor prescribing, and GLP-1 adoption
- Supporting pre-launch analytics for investigational denecimig (Mim8), informing HCP targeting and field readiness

ZS Associates

Pune, IN

Advanced Data Science Associate Consultant

Feb 2025 - Jun 2025

- Built and deployed an org-wide analytics and ML platform (Spark/SQL, dashboards) that unified 5+ data sources into territory and product KPIs used by 100+ stakeholders supporting a \$10B oncology portfolio
- Cut weekly reporting time from days to minutes, replacing Excel workflows with automated pipelines and self-serve dashboards

ZS Associates

Pune, IN

Decision Analytics Associate Consultant

Jul 2024 - Jan 2025

- Led a 5-member team to modernize legacy business rules; saved ~50 hrs/mo and improved first-pass quality to >99%
- Built and deployed Positive-Unlabeled learning models to infer missing categorical labels in medical transaction data, increasing customer-journey analytics coverage from ~40% to ~95% across tumor types and territories
- Implemented drift monitoring (feature + prediction drift) and CI unit tests for production pipelines, reducing silent failures
- Placed in the top ~10% in a company-wide hackathon and earned selection for a lateral transfer into the Data Science vertical

ZS Associates

Pune, IN

Decision Analytics Associate

Feb 2022 - Jun 2024

- Engineered PySpark/SQL ETL pipelines integrating multiple healthcare data sources covering millions of patients for \$4B+ oncology drug performance analytics
- Defined patient cohort inclusion and exclusion logic robust to missing and miscoded fields, enabling audit-ready reporting
- Delivered ad hoc analyses identifying care gaps and market opportunities to inform brand strategy across multiple launches
- Promoted to Associate Consultant in 4 cycles (typical: 5) and received Expert Associate and Insight Illuminator awards

PROJECT EXPERIENCE

DEUCE: Tennis Forecast Engine (ATP + WTA) - Python, XGBoost, Next.js, GitHub Actions

[GitHub](#) / [Live Site](#)

- Built a hybrid forecast engine over 88K+ tour matches combining surface Elo with cross-surface transfer, an opponent-adjusted serve/return Markov point model, and a Platt-calibrated XGBoost combiner with Monte Carlo draw simulation
- Beat the bookmaker benchmark on walk-forward, leakage-free ATP predictions (0.195 vs 0.196 Brier, 69.6% vs ~69% accuracy); ships with hourly data refresh, daily retrains, and 182 CI tests via GitHub Actions

GAFFER: FIFA World Cup 2026 Forecast Engine - Python, Next.js, Firebase

[GitHub](#) / [Live Site](#)

- Modeled all 48 teams using Elo ratings from 150 years of international results blended with squad market value, a Dixon-Coles goal model, and 50,000 Monte Carlo tournament simulations; live odds auto-refresh as matches complete

Airbnb Data Analyst Agent - Python, LangChain, FastAPI, DuckDB, Next.js

[GitHub](#) / [Live Demo](#)

- Built a multi-agent data analyst (Planner, SQL, Validator, Chart, Narrator agents on a typed message bus) turning natural-language questions into validated SQL, charts, and narrated answers with row-level citations
- Engineered guardrails (SQL dry-runs, row-count and type audits, self-critique with retries) plus a golden Q/A regression eval suite and per-query latency and cost tracking; pluggable DuckDB, Postgres, and Snowflake backends

Biliary Tract Cancer (BTC) early detection using ML - Python, scikit-learn, PySpark

- Developed an early detection model across 250M patient claims using clinical risk factors, K-means/GMM segmentation, and Transformer-based NLP clustering, enabling ~45-day earlier identification vs standard diagnosis lag
- Presented methodology at PMSA 2025 conference; model adopted for territory-level resource planning

Financial RAG Chatbot - Python, FastAPI, ChromaDB, Streamlit, GCP

[GitHub](#)

- Built an LLM-powered RAG chatbot answering financial questions from SEC filings with line-level citations, using ChromaDB semantic retrieval, text-embedding-3-large, and automatic ticker and period parsing
- Built a multi-model evaluation pipeline using Claude Opus as a judge and deployed a live demo on Streamlit Cloud

Agricultural Product Standardization & Risk Detection - SunCulture (Series B Agtech) - Python, RAG, REST API

- Built a RAG-augmented classification system categorizing 7M+ farmer transactions across 500+ product categories to support creditworthiness assessment for microloans in East Africa
- Achieved 99% accuracy on a 10K-item holdout via hybrid rule-based + LLM-assisted classification, cutting manual review by 95%

SKILLS

Programming: Python, SQL, R, C++

Analytics & ML: classification, regression, NLP, clustering, model evaluation, LLMs/RAG, prompt engineering

Data & MLOps: Spark (PySpark), Databricks, ETL/ELT, MLflow, Docker, AWS (S3, EMR, Athena, SageMaker)

Libraries: pandas, NumPy, scikit-learn, PyTorch, SHAP, matplotlib, BeautifulSoup

Tools: Git, Linux, CI/CD, Jupyter, Jira, Confluence, Claude Code, Cursor IDE